

Body systems: the dispatch challenge

Teacher guide and worked answers

Describe digestion and circulation, then explain how they work together.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | B32:C32

Describe the main human body systems (circulation, digestion).

Directly develops this stated outcome with secondary-age tasks at an upper-KS2 conceptual level. Support and stretch are flexible; this exemplar is not a claim of qualification approval.

Prior learning

Name the heart and stomach.

Know that food contains substances the body uses.

Success evidence

Match organs to the digestive or circulatory system.

Explain that digestion breaks food down and circulation transports substances.

Repair a model that sends whole food into blood vessels.

Equipment

Shared evidence sheet and selected route; pencils; optional reusable organ/job cards.

Preparation

Print one shared evidence sheet per pupil; cut cards only if useful.

Preview the two-system model. It is a job map, not an anatomical picture.

Practical care

No food, bodily samples, exercise or personal health measurements are needed.

Allow pupils to use labels rather than draw an outline of their own body.

Ready to teach alternative

All work is a paper or screen model using supplied organ/job cards.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A delivery puzzle
2	Retrieval	2-5	Name it, then say its job
3	I do	5-13	I follow the food route

Slide	Stage	Minutes	Focus
4	I do	Within stage	I follow the transport route
5	I do	Within stage	I connect the two systems
6	We do	13-21	Organ-and-job dispatch
7	We do	Within stage	The mixed-up display
8	We do	Within stage	Check the delivery claim
9	Hinge	21-24	Which explanation links the systems?
10	You do	24-34	Build the museum panel
11	You do	Within stage	A useful sentence frame
12	You do	Within stage	Proofread for science
13	Review	34-38	Museum quality check
14	Review	Within stage	Choose the next question
15	Exit	38-40	Leave a clear explanation

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A delivery puzzle

Give 20 seconds to think, draw or point. Do not require personal food talk. Say: We are tracing a fictional lunch, not judging anyone's choices.

2 Retrieval Name it, then say its job

R1 heart or lungs; R2 mouth, stomach or intestine. R3 no; a job statement is also needed. Accept familiar language and supply scientific labels.

3 I do I follow the food route

Think aloud: I place mouth before stomach because food enters at the mouth. I add the small intestine because digestion continues and most nutrients are absorbed there. This is a selected job map; it does not show every organ or their shapes.

4 I do I follow the transport route

Shared I do time. Trace clockwise. Think aloud: The system keeps blood moving; it does not chew or digest food. This simplified loop leaves out the lung route, which we investigate next lesson. It is not a complete circulatory route.

5 I do I connect the two systems

Shared I do time. Model correction: "The heart sends sandwiches to muscles" becomes "Small absorbed nutrients are carried in blood to tissues." Digestion and circulation are distinct but linked. Teacher nuance: most dietary fat first enters lymph; this core model follows a soluble nutrient such as glucose.

6 We do Organ-and-job dispatch

Work in pairs or independently with pointing. Read each card once. Mouth, stomach, small intestine digestive; heart and blood vessels circulatory. Ask pupils to justify a disputed placement. Cards are on pupil page 1.

7 We do The mixed-up display

Shared We do time. Invite pupils to move labels or read a repaired statement. Answer: stomach mixes food with digestive juices; heart pumps blood. A vague “stomach eats” needs refining.

8 We do Check the delivery claim

Shared We do time. Reject as written; digested small nutrients, for example glucose, can be absorbed and transported by blood. Praise correct links, not speed.

Reveal: Small absorbed nutrients travel in blood; whole food pieces do not.

9 Hinge Which explanation links the systems?

A: confuses jobs. B correct and connects both. C: organs do not work alone. Re-teach with two cards, one for processing and one for transporting.

A. The heart digests lunch. - The heart pumps blood; digestive organs digest food.

B. Digestion produces small nutrients; blood transports them. - This gives the two different, connected jobs.

C. The stomach works without any other body system. - Systems work together; tissues also need a blood supply.

10 You do Build the museum panel

10 minutes across three slides. All routes assess organ jobs and the connection. A drawing with dictated labels is valid independent evidence.

11 You do A useful sentence frame

Shared You do time. Prompt by asking what happens to food and what moves blood. Do not fill the gaps for pupils.

12 You do Proofread for science

Shared You do time. Check pupils do not label whole-food arrows into blood. Stretch should name a limitation: omitted organs, approximate positions or omitted lymph route.

13 Review Museum quality check

4 minutes across Review. Offer private teacher sharing. Model feedback: Your heart label is right; now explain what carries nutrients away.

14 Review Choose the next question

Lundy: collect a pointed, spoken or written question. Teacher chooses one question about the heart's route for next lesson and tells pupils how their suggestion shaped the starter.

15 Exit Leave a clear explanation

Heart pumps blood. Digestive organ with appropriate job. Digestion provides small absorbable nutrients that blood carries to tissues. Collect before public answers.

Answers and assessment

R1-R3

R1 heart or lungs. R2 stomach, mouth or an intestine. R3 no; a name identifies the part, whereas a job statement explains what it does.

W1-W3

W1 A/B/C digestive; D/E circulatory, with job evidence. W2 stomach mixes food with digestive juices; heart pumps blood. W3 replace bread crumbs with small absorbed nutrients, for example glucose.

Hinge A-C

A incorrect: heart pumps. B correct: different connected jobs. C incorrect: systems cooperate. Re-teach by tracing one food route and one blood route.

S1-S5

S1 digestive. S2 pumps blood. S3 A → B → C; accept correct job. S4 absorbed; transports. S5 a named digestive organ with correct job, heart pumps blood and an arrow for small absorbed nutrients carried in blood; whole food into blood is incorrect.

N1-N4

N1 mouth/digestive/chews; small intestine/digestive/digests and absorbs; heart/circulatory/pumps; vessels/circulatory/carry blood. N2 mouth → stomach → small intestine is an acceptable selected route; full route may include oesophagus. N3 small substances produced during digestion cross the gut wall and are carried in blood to muscle tissue. N4 food pieces remain in the gut; small absorbed nutrients travel in blood.

T1-T4

T1 must distinguish digestion and circulation and explicitly connect absorption to transport. T2 add other digestive organs such as mouth and small intestine; replace pumping role with heart and vessels. T3 valid omissions include oesophagus, large intestine, lung route or detailed absorption. T4 an objective-aligned question such as “Does the heart digest food? Explain” with a correct expected response. Accept other accurate questions.

E1-E4 / reflection

E1 pumps blood. E2 valid organ/job pair. E3 small absorbed nutrients from digestion are carried in blood to tissues. E4 testable or explanatory heart question, for example “Where does blood go after the heart?” Reflection has no prescribed preference.

Assessment and re-teach

Secure: identifies at least one organ per system, gives correct jobs and explains the absorption-transport connection. If only organ names are secure, require one verb per organ. If whole-food transport persists, contrast food pieces with small dissolved nutrient particles.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This lesson establishes system roles; the following heart lesson develops the double circulation, while the later transport lesson examines crossing the gut wall.

The job diagrams deliberately omit detailed anatomy and the complete circulatory route. Name that limitation. Do not teach that all dietary fats pass directly into blood: most enter the lymphatic system first. Interactive We do uses predict, test and explain; paper cards W1-W3 provide the same decisions. A desk token can trace the model without an animated screen. Movement is optional and pupils control the pace.

Access and responsive teaching

Supported

Match organ and job cards; use two sentence frames.

Standard

Build and explain a two-system job map.

Stretch

Repair a misleading museum display and explain the model boundary.

Regulation

Offer quiet pointing, private writing, drawing or adult scribing. Pass-and-return for public questions. Use fictional cases; no personal medical disclosure or comparison of bodies. Assign response routes from current evidence, not fixed pupil labels.

Misconceptions to check

The heart digests food.

The digestive system breaks food down; the heart pumps blood.

Check: Ask which organ squeezes to pump blood.

Pieces of sandwich travel around blood vessels.

Small absorbed nutrients travel in blood; recognisable food pieces stay in the digestive tract.

Check: Ask whether a blood vessel contains a whole bread crumb.

One organ does every job in a system.

Several organs contribute different jobs to the same system.

Check: Ask why mouth and small intestine are both needed.

Word	Meaning
organ	A body part made of tissues that carries out particular jobs.
system	Organs working together to carry out major jobs.
digestion	Breaking food into smaller substances that can be absorbed.
circulation	The movement of blood through the heart and blood vessels.
nutrient	A substance the body needs for energy, growth or other functions.

Scientific references

NIDDK: [Your digestive system and how it works](#)

Checked September 2026. Digestion, absorption, transport and water absorption. Paraphrased factual reference; diagrams and activities are original.

NHLBI: [How blood flows through the heart](#)

Checked September 2026. Right-heart/lung/left-heart/body sequence and oxygenation. No adult pulse thresholds are used for pupils.